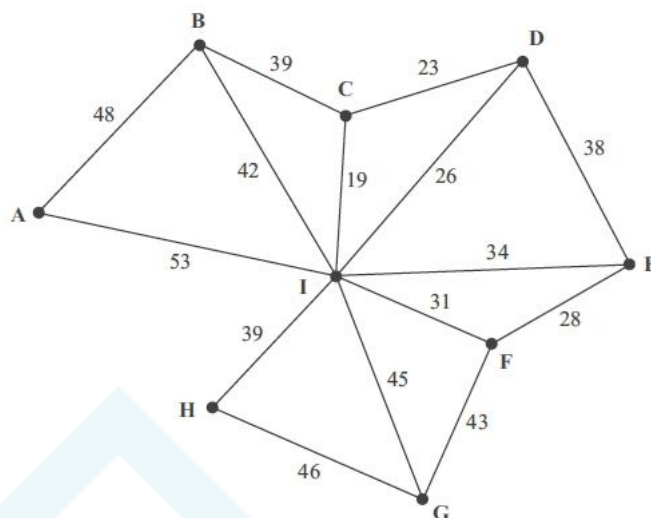


Given: Due:	Further Maths (Decision) HW 10	MARK SCHEME
Chapter 3: Algorithms on graphs (MST and shortest paths)		
1) (a) Use Kruskal's algorithm to find a minimum spanning tree for the network shown in Figure 2. You should list the arcs in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree. (3) (b) Starting at A, use Prim's algorithm to find a minimum spanning tree for the network in Figure 2. You must clearly state the order in which you include the arcs in your tree. (3) (c) Draw a minimum spanning tree for the network in Figure 2 using the vertices given in Diagram 1 of the answer book. State the weight of the minimum spanning tree. (2) A new spanning tree is required which includes the arcs DI and HG, and which has the lowest possible total weight. (d) Explain which algorithm you would choose to complete the tree, and how the algorithm should be adapted. (You do not need to find the tree.) (2)		



Around 25% of the candidates scored full marks on this question. In part (a) a number of candidates only stated the arcs they were including in their tree and did not state the arcs that they rejected, as they rejected them. Some candidates only referred to the length of the arc rather than by its end vertices, this makes it difficult for the examiners to determine which arc is being considered. Part (b) was usually completed correctly, although some candidates referred to arcs that they had rejected. Some candidates wasted time drawing a table to run Prim, and then showed their working on the table. If they listed the arcs in order they gained full credit, but some only listed the nodes in order. Many candidates knew that Kruskal needed to be used, though some were unclear about how they would modify it. Incorrect answers included Prim, Dijkstra, Route Inspection and Hamiltonian cycle.

Question Number	Scheme	Marks
(a)	CI CD (not DI) EF FI (not EI not DE) $\begin{Bmatrix} BC \\ HI \end{Bmatrix}$ (not BI) GF (not GI not HG) AB	M1 A1 A1 (3)
(b)	AB BC CI CD FI EF IH FG	M1 A1 A1 (3)
(c)	 Weight: 270	B1 B1 (2)
(d)	Start off the tree with DI and HG and then apply Kruskal's algorithm	B2, 1, 0 (2) [10]
Notes		
(a)	1M1: Kruskal's algorithm – first 4 arcs selected chosen correctly. 1A1: All eight non-rejected arcs chosen correctly. (Working seen in (a)) 2A1: All rejections correct and in correct order and at correct time.	
(b)	1M1: Prim's algorithm – first four arcs chosen correctly, in order, or first five nodes chosen correctly, in order. {A, B, C, I, D} (arcs not arc lengths) 1A1: First six arcs chosen correctly; all 9 nodes chosen correctly, in order. {A, B, C, I, D, F, E, H, G} [1 2 3 5 7 6 9 8 4] 2A1: cso	
(c)	1B1: cao (condone lack of numbers) 2B1: 270 cao	
(d)	1B1: Kruskal's algorithm + some argument 2B1: Kruskal's algorithm + start with the two arcs. (o.e)	

2) Figure 1 represents the distances, in km, between eight vertices, A, B, C, D, E, F, G and H in a network.

(a) Use **Kruskal's** algorithm to find the minimum spanning tree for the network. You should list the arcs in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree.

(3)

(b) Starting at A, use **Prim's** algorithm to find the minimum spanning tree. You must clearly state the order in which you selected the arcs of your tree.

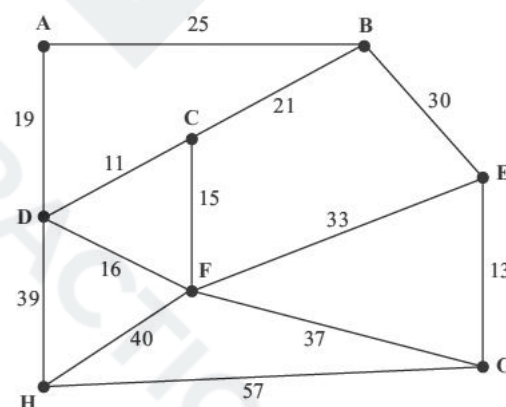
(c) Draw the minimum spanning tree using the vertices given in Diagram 2 in the answer book.

(3)

(1)

(d) State the weight of the tree.

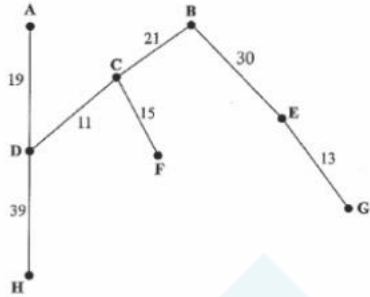
(1)



This proved a good starter with 67% of the candidates gaining full marks and only 20% gaining 6 or fewer marks. The most efficient way of applying Kruskal's algorithm was to list all the arcs in order then use ticks and crosses to show the selection of the arcs used to form the tree. A few candidates rejected BE or DH and added EF or FG. Part (b) was similarly well done. However a number of candidates spent time (unnecessarily) constructing a matrix; others made reference to arcs being rejected (this should not happen if Prim's algorithm is applied correctly). It was noticeable the fewer candidates listed only the vertices. Most candidates were able to draw the tree and state the weight correctly. It is true that both algorithms should



give rise to MST's that have the same weight. This could be a useful check for some candidates.

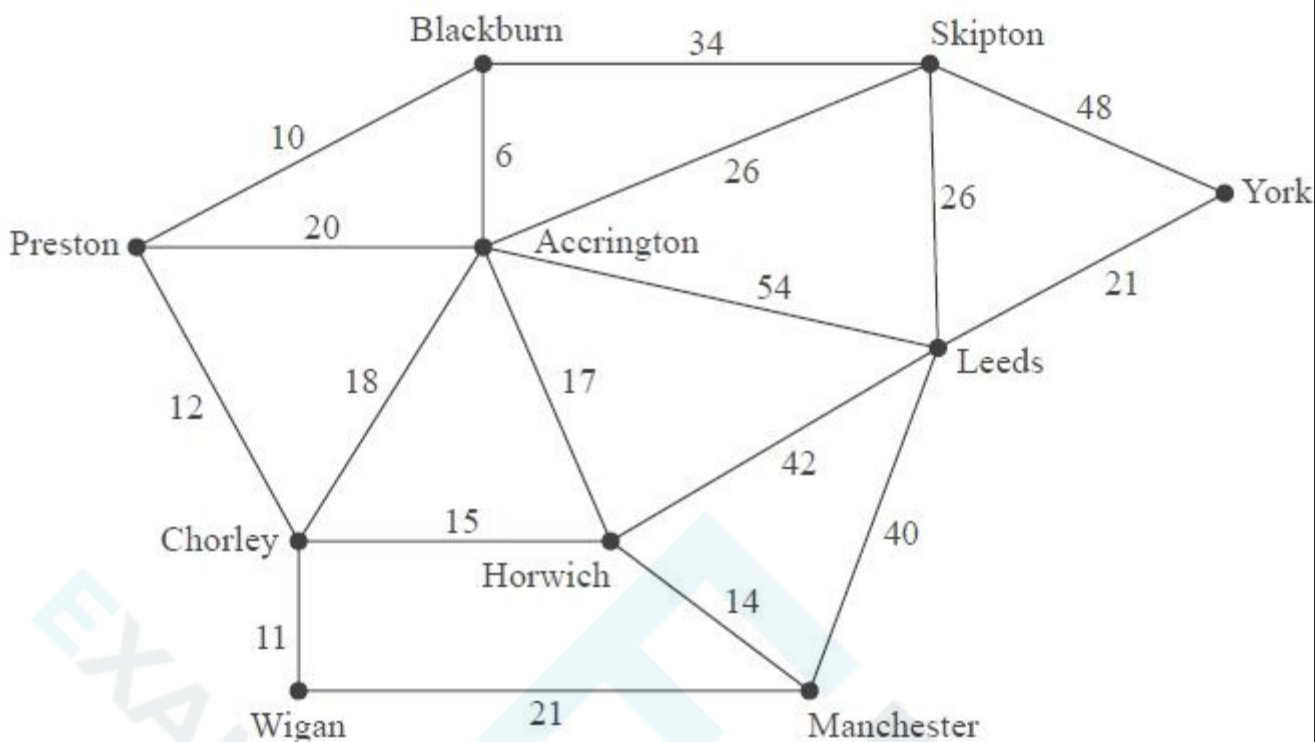
Question Number	Scheme	Marks
(a)	DC, EG, CF, reject DF; AD, BC, reject AB, BE, reject EF and FG, DH	M1; A1 A1 3
(b)	AD, DC, CF, CB; BE; EG, DH	M1; A1; A1 3
(c)		B1 1
(d)	Weight of tree = 148 (km)	B1 1 Total 8

Notes

- a1M1: First three arcs correctly chosen and DF rejected. Accept weights for all 3 marks.
 Special case: If all 7 arcs, in correct order, but no rejections seen at all, then award M1 only.
- a1A1: All arcs/weights in tree selected correctly at correct time.
- a2A1: All rejections correct and at the right time.
- b1M1: First four arcs/weights correctly chosen, or first five nodes ADCFB chosen in order.
 Special case : If Prim but not starting at A please send to review.
- b1A1: First five arcs/weights correctly chosen, or all nodes in order A, D, C, F, B, E, G, H.
- b2A1: CSO (must be arcs/weights). E.g no 'reject' arcs
- c1B1: CAO mark what you see at (c).
- d1B1: CAO mark what you see at (d).



3)



Sharon is planning a road trip from Preston to York. Figure 2 shows the network of roads that she could take on her trip. The number on each arc is the length of the corresponding road in miles.

(a) Use Dijkstra's algorithm to find the shortest route from Preston (P) to York (Y). State the shortest route and its length.

(6)

Sharon has a friend, John, who lives in Manchester (M). Sharon decides to travel from Preston to York via Manchester so she can visit John. She wishes to minimise the length of her route.

(b) State the new shortest route. Hence calculate the additional distance she must travel to visit John on this trip. You must make clear the numbers you use in your calculation.

(3)

Question (a) was usually very well done with most students applying Dijkstra's algorithm correctly. The boxes at each node in question (a) were usually completed correctly. When errors were made it was either an order of labelling error (some students repeated the same labelling at two different nodes) or working values were either missing, not in the correct order or simply incorrect (usually these errors occurred at nodes S, L and/or Y). The route was usually given correctly and most students realised that whatever their final value was at Y this was therefore the value that they should give for their route.

Question (b) was done well although many students failed to read the question carefully and simply calculated the length of the new shortest route without stating the increase in the distance travelled. Some failed to realise that the final value at vertex M added to the length of the arcs ML and LY would give them the required length of the new shortest route, instead many added up the length of each arc from scratch.



Question Number	Scheme	Marks
(a)		M1 A1 (PBCAW) A1 (HMS) A1ft (LY)
	Shortest route: P – B – A – S – L – Y Length: 89 (miles)	B1 B1ft (6)
(b)	Shortest route: P – C – H – M – L – Y Difference in routes: $(41 + 40 + 21) - 89 = 13$ (miles)	B1 M1 A1 (3) 9 marks

Notes for Question

In (a) it is important that all values at each node are checked very carefully – the order of the working values must be correct for the corresponding A mark to be awarded e.g. at L the working values must be 70 69 68 – in that order (70 68 69 is incorrect). The values in brackets in the working values at P, A, H and L can be ignored but if a candidate does have additional values at these nodes then they must be these ones only. Penalise any other/incorrect working values with the corresponding A mark. It is also important that the order of labelling is checked carefully – some candidates start with a label of 0 at P (rather than 1) – this is fine. Also the order of labelling must be a strictly increasing sequence – so 1, 2, 3, 3, 4, ... will be penalised once (see notes below) but 1, 2, 3, 5, 6, ... is fine.

a1M1: A larger value replaced by smaller value at least once in the working values at either A or M or L or S or Y.

a1A1: All values in P, B, C, A and W correct. The working values at A must be in the correct order. Condone lack of 0 in P's working value. Ignore additional working value of 30 at the end of A (may read 20 16 30 – rather than 20 16 – at A).

a2A1: All values in H, M and S correct and the working values in the correct order. Penalise order of labelling only once per question (H, M and S labelled in that order and H must be labelled after P, B, C, A and W). Ignore additional working value of 33 at the end of H (may read 27 33).

a3A1ft: All values in L and Y correct on the follow through and the working values in the correct order. Penalise order of labelling only once per question (L and Y labelled in that order and L labelled after all other nodes (excluding Y)). Ignore additional working value of 81 at L – may read 70 69 81 68 – rather than 70 69 68 – which is fine – however, 70 69 68 81 is incorrect and loses this mark.

To follow through L check that all the working values at L follow from the candidate's final values from nodes A, H, M and S (in whatever order the candidate has labelled these four nodes) and that the final value, and order of labelling, follows through correctly. Repeat for Y (which will have working values from S and L).

a1B1: CAO for the route (or starting at Y to P).

a2B1ft: Follow through on their final value at Y – if their answer is not 89 follow through their final value at Y (condone lack of units).

b1B1: CAO for the route (or starting at Y to P).

b1M1: Their final value at M + 40 + 21 – accept a value of 102 (with no other working) for this mark.

b1A1: CAO (condone lack of units) – accept, as a minimum, 102 followed by 13 for both marks. If 13 with no working then award the previous M mark but withhold the final A mark.